

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2018 - 2019 (Even Semester)

Innovative Practices Description

UNIT V MULTIPLE ANTENNA TECHNIQUES

Degree, Semester& Branch: VIII Semester B.E. ECE B

Course Code & Title: EC6801 Wireless Communication

Name of the Faculty member: Mrs.G.Gnana Priya

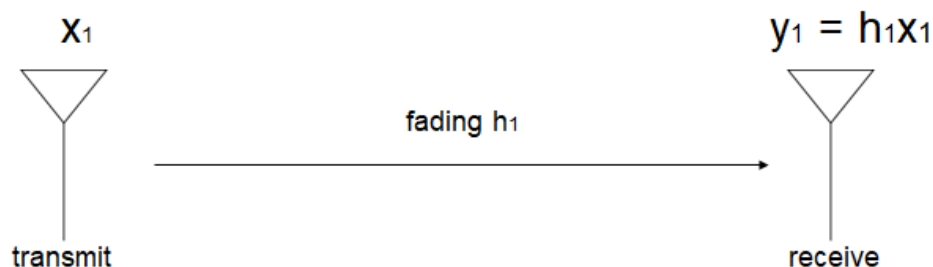
Name of the Topic: SIMO, MISO, SISO and MIMO systems

Name of the Innovative Practice: Power Point Presentation

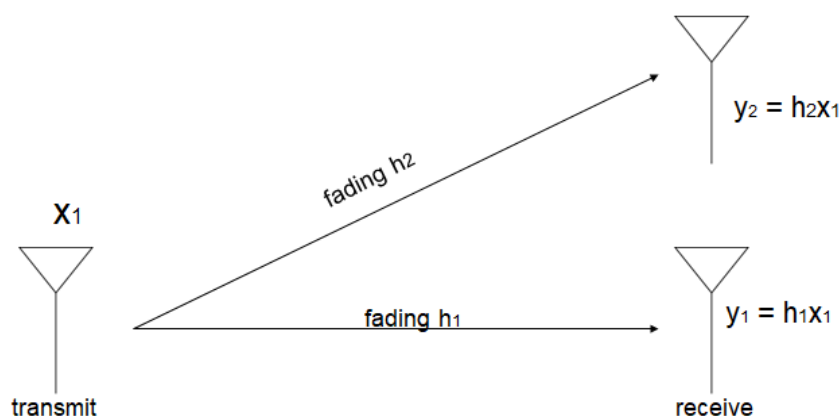
Description:

In wireless communications, there is always a need for increase in performance in wireless systems with significant increase in spectral efficiency and data rates, high Quality of Service (QoS) ,wide coverage, etc., A detailed explanation about the comparison among SISO(Single Input Single Output), SIMO (Single Input Multiple Output), MISO (Multiple Input Single Output) and MIMO (Multiple Input Multiple Output) was shown to the students.

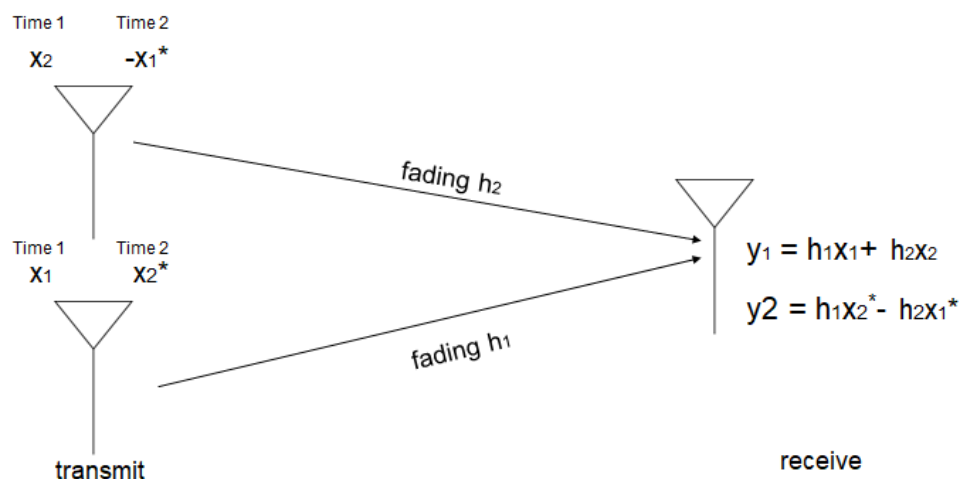
SISO (Single Input Single Output):



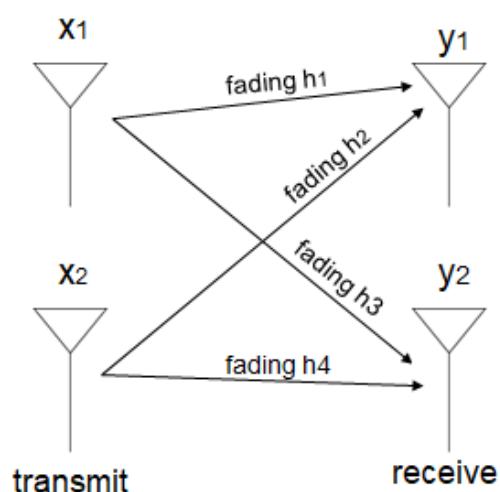
SIMO (Single Input Multiple Output):



MISO (Multiple Input Single Output):



MIMO (Multiple Input Multiple Output):



Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	2	2	3		2			1		2		2

CO5: The students will be able to implement the concept of transmit/receive diversity in MIMO systems

References:

- Rappaport, T.S, "Wireless communications", Second Edition, Pearson Education, 2010.
- Andreas.F. Molisch, "Wireless Communications", John Wiley – India, 2006.